YAT758_ONL5_DAT3_G5_DEPI5 **Group 3**

Project Name: **Store Sales**

Git Hub Repository link:

https://github.com/marwahussein-hub82/Store-Sales-Group3

Project Idea: **Store Sales**

This project turns four years of retail order data into business insights, sales forecasts and an interactive reports & dashborads that helps decision makers plan sales, products and shipping.

Project Description:

The dataset covers a US retail store from January 2015 to December 2018.

- **Size:** 9,800 order lines, 4,922 orders, 793 customers, 1,861 products, 49 states.
- **Total sales:** about $2.26 million.
- **Main fields:** order and ship dates, ship mode, customer and segment, location (region, state, city), product (category, sub-category, name) and sales amount.

Project Goal:

Give decision makers a clear view of what drives sales and what to expect next, through clean data, evidence-based analysis, accurate forecasts, reports and a simple dashboard.

Objectives:

1. Build a clean, well-structured data model ready for analysis.
2. Find the best and worst performing categories, regions, segments, customers and periods.
3. Measure the combined impact of product category and region on sales.
4. Forecast sales for the next 6–12 months.
5. Deliver dashboard, a final report and a presentation with practical recommendations.

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Project Team Members:

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5.	Mohamed Salah Ammar	

Work Plan:

Week	Phase	Deliverables
1	Data model, cleaning and preprocessing	Cleaned dataset and preprocessing notebook
2	Analysis questions	Set of analysis questions with answers
3	Forecasting questions	Forecast visualization
4	Dashboard and final presentation	Dashboard, report and presentation

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Roles & Responsibilities:

We are working on all the tasks “grouping” according these roles

Team Member	Role	Week 1	Week 2	Week 3	Week 4
	Data Engineer	Designs the SQL data model, loads the data and writes the validation queries	Supports SQL queries for all questions	Prepares monthly time-series datasets	Creates the SQL views and tests the data connection
	Data Analyst (Products and Regions)	Cleans the data in pandas (dates, missing values, duplicates, product names)	Answers questions	Builds the Linear Regression baseline and the category forecasts	Writes the report sections on data, cleaning and analysis
	Data Analyst (Customers and Operations)	Creates derived features (Year, Month, Quarter, Shipping Days) and quality charts	Answers questions	Builds the Random Forest or Gradient Boosting model, evaluates MAE, RMSE and MAPE, and forecasts by region	Writes the report sections on forecasting, recommendations and limitations
	BI Developer and Presentation Lead	Documents the data dictionary and the before/after cleaning summary	Reviews the charts and keeps the question document consistent	Produces the final forecast plots and interprets the results	Builds and tests the dashboard and prepares the final presentation

Tools:

SQL, Python, Tableau, Power BI

Instructor: Eng. Abdelrahman Ashour